Eulalia VoterID System — Proof of Concept Results

1. Executive summary

Eulalia VoterID is a proof of concept (PoC) for digital voter identity and political affiliation. The project validates a flow in which a citizen can interact with a web or mobile client, use backend services, and progress through a self-sovereign identity (SSI) process before political affiliation is accepted. The PoC demonstrates technical integration across the main application components and Identus/Atala PRISM infrastructure. It does **not** claim production readiness, electoral certification, or public deployment.

2. Project description and objectives

The project explores how digital identity, biometric-related validation, and SSI credentials can support voter identity and political affiliation workflows.

Objectives:

- Provide backend endpoints for citizens, authentication, political affiliation, biometrics, SSI, and organization requests.
- Provide a web interface and a Flutter mobile application for interacting with the PoC.
- Validate the integration of an Identus/Atala PRISM Cloud Agent with the application flow.
- Exercise the main registration, authentication, invitation, webhook, status, credential, and affiliation gates.
- Produce reviewable technical and test evidence without presenting the PoC as a production election system.

3. Methodology and activities

The work followed an iterative implementation and validation approach:

1. Implement and connect backend APIs, web views, and mobile flows.
2. Configure the SSI infrastructure and verify Cloud Agent availability.
3. Execute component tests, API integration tests, and SSI end-to-end validation.
4. Record results, limitations, and evidence in the project documentation repository.

The SSI flow was documented as: invitation generation, webhook processing, status consultation, and credential issuance. Political affiliation is subject to validation gates, including the required SSI status and biometric-related conditions.

4. Implementation and architecture

Layer	PoC implementation
Backend	.NET 8 with PostgreSQL; APIs for citizens, authentication, political affiliation, biometrics, SSI, and organization requests
Web client	React with Vite
Mobile client	Flutter application
SSI infrastructure	Identus/Atala PRISM Cloud Agent and supporting Docker services
Evidence	Test and integration evidence maintained in `eulalia-documents/integration-validation`

The Identus/Atala PRISM evidence includes a Docker stack with APISIX, Cloud Agent, PRISM node, PostgreSQL, Vault, and Swagger UI. The health endpoint `/cloud-agent/_system/health` returned Cloud Agent version `1.39.0`. This verifies the tested environment; it is not evidence of a public production deployment.

5. Results

The PoC achieved the following validated outcomes:

- Backend, web, mobile, and SSI infrastructure components were exercised together.
- Citizen registration and authentication API paths were validated.
- SSI invitation, webhook, status, and credential issuance steps were documented and tested.
- The application enforces a security gate for political affiliation: the SSI credential status must be suitable for the operation, together with the applicable base and biometric checks.
- The web test set reported 3 of 3 tests passed.
- The Flutter test set reported 1 of 1 test passed.
- The SSI end-to-end validation reported 5 of 5 tests passed.

6. Testing and validation

The consolidated test report records **147 tests overall: 144 passed and 3 failed or conditional, reported as 98.0%**. The figures must be read with the limitations below.

Area	Result	Limitation or interpretation
Backend .NET	132 total; 130 passed; 2 failed	Backend coverage was 12.4% line and 25% branch coverage
React/Vite frontend	3/3 passed	The reported set is small and does not represent complete product coverage
Flutter mobile	1/1 passed	The reported set is small and does not represent complete product coverage
API integration	6 tests; 5 passed; 1 conditional	The affiliation validation returned a conditional result because the required credential state was not available in validation that test path
SSI end-to-end	5/5 passed	Validates the documented PoC flow, not production operations

The two backend failures and the conditional API result remain explicit limitations. The aggregate percentage must not be interpreted as a production-readiness or security certification metric.

7. Challenges and solutions

Challenge	Response in the PoC
SSI operations depend on ordered state transitions	The flow exposes invitation, webhook, status, and credential stages and applies validation gates before affiliation
Integration spans several services	The implementation separates backend, clients, and Identus infrastructure so each can be tested and evidenced independently
Test coverage is uneven	Component results and coverage limitations are disclosed rather than generalized into a readiness claim

Affiliation depends on credential state	The conditional validation is retained as evidence of the security gate and as a follow-up item

Platform transition and milestone constraints

The initial project baseline referenced Atala PRISM. During execution, the SSI platform baseline changed because the original Atala PRISM path was no longer available or maintained for this project, requiring integration and validation with Identus. This required adaptation of the integration work, evidence, terminology, and validation scope; therefore, some deliverables should be interpreted against the revised baseline rather than the initial plan.

The journal article/publication deliverable was affected by this transition and by other pre-established milestone conditions. This is documented as a delivery constraint affecting the planned commitments, not as a failure or direct rejection. The final evidence demonstrates the revised Identus-based PoC and does not claim full equivalence with every original Atala PRISM commitment.

8. Lessons learned

- A PoC must distinguish an integrated demonstration from a production deployment.
- SSI state transitions are part of the business validation, not merely an infrastructure detail.
- Security gates can cause an expected conditional API result when prerequisite credential state is absent.
- Evidence links and reproducible test descriptions are essential for reviewer confidence.
- Small passing client test sets do not replace broader service, widget, integration, performance, and security testing.

9. Recommendations

1. Resolve the two failing backend tests and preserve regression coverage for the validation rules.
2. Expand backend line and branch coverage beyond the current 12.4% and 25% results.
3. Add broader React, Flutter, service, widget, API, and end-to-end tests.
4. Validate biometric workflows with controlled test data and document their security assumptions.
5. Conduct threat modeling, privacy review, operational monitoring, load testing, and deployment review before any pilot or production consideration.
6. Define governance, credential revocation, recovery, data retention, and electoral compliance requirements.

10. Conclusion

Eulalia VoterID demonstrates a technically integrated PoC for digital voter identity and political affiliation using .NET, PostgreSQL, React/Vite, Flutter, and Identus/Atala PRISM components. The documented tests provide evidence for selected flows, while the failures, conditional validation, limited coverage, and absence of a public production deployment remain material limitations. Further engineering and institutional validation are required before the system could be evaluated for a real-world pilot.

11. Relevant links

- Backend (https://github.com/democracyonchain/eulalia-backend)
- Web frontend (https://github.com/democracyonchain/eulalia-frontend)
- Flutter application (https://github.com/democracyonchain/eulalia-app)
- Atala PRISM / Identus integration (https://github.com/democracyonchain/eulalia-atala-identus)
- Project documentation (https://github.com/democracyonchain/eulalia-documents)
- Integration-validation evidence (https://github.com/democracyonchain/eulalia-documents/tree/main/integration-validation)
- Test reports (https://github.com/democracyonchain/eulalia-documents/tree/main/tests/TestIntegration)

Milestone scope note: The closing video and magazine article are separate milestone deliverables and are not covered by this report. Publication was also affected by the platform transition and other pre-established milestone conditions; this report therefore keeps the publication deliverable out of scope while documenting the constraint.